# Risk Alert Classifier — Final Report

Digital-banking early-warning system for high-risk customers · Rensee Gajipara

## Dataset & approach

4,600 customers, 17 usable features (demographic, transaction and credit-behaviour), target *risk_status* (12.1% High Risk). Stratified 80/20 split; KNN imputation (k=5) for 7 numeric/categorical columns with missing values; Logistic Regression baseline; Under/Over/SMOTE/ADASYN; Decision Tree and Random Forest; RandomizedSearchCV + GridSearchCV; ROC analysis; threshold chosen for minimum false negatives.

## Imbalance techniques (Logistic Regression)

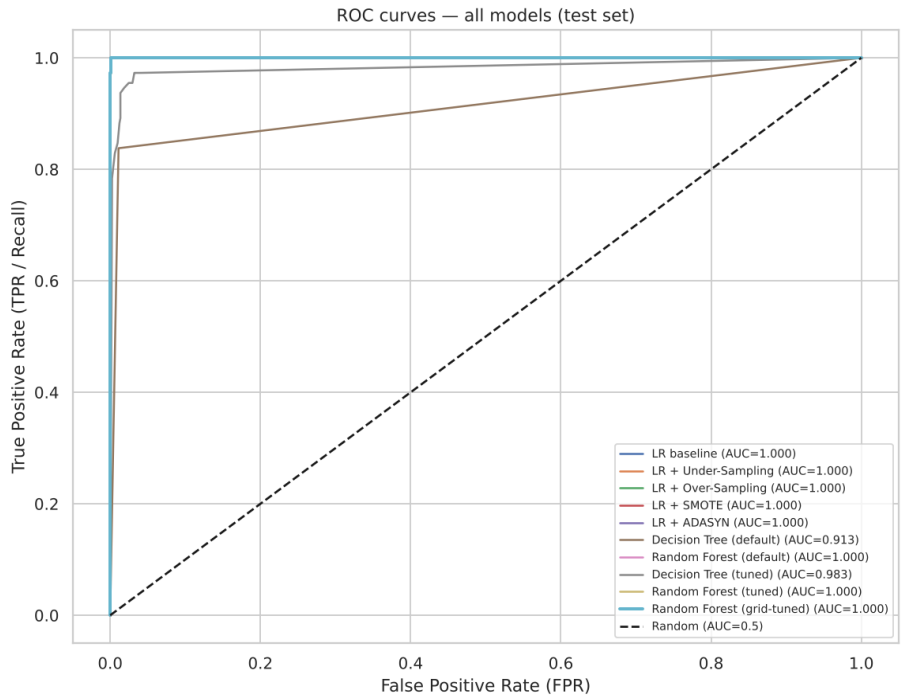
| Model | Accuracy | Precision | Recall | F1 | AUC | FP | FN |
|---|---|---|---|---|---|---|---|
| LR + No balancing | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 0 | 0 |
| LR + Under-Sampling | 0.998 | 0.982 | 1.000 | 0.991 | 1.000 | 2 | 0 |
| LR + Over-Sampling | 0.998 | 0.982 | 1.000 | 0.991 | 1.000 | 2 | 0 |
| LR + SMOTE | 0.998 | 0.982 | 1.000 | 0.991 | 1.000 | 2 | 0 |
| LR + ADASYN | 0.996 | 0.965 | 1.000 | 0.982 | 1.000 | 4 | 0 |

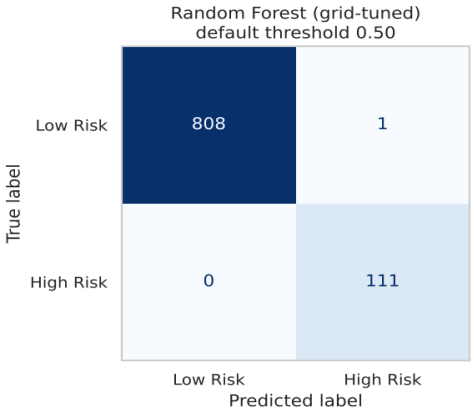
## Tuned vs untuned tree models (test set)

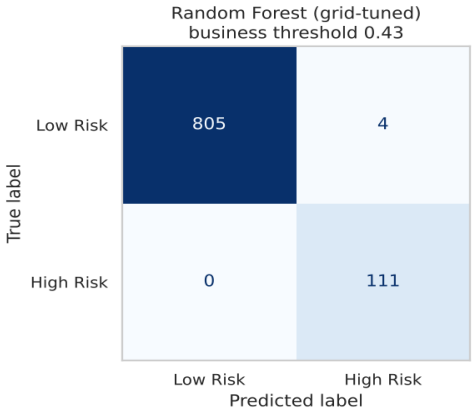
| Model | Train_F1 | Test_Acc | Precision | Test_Recall | Test_F1 | Test_AUC | FN | FP |
|---|---|---|---|---|---|---|---|---|
| Decision Tree (default) | 1.000 | 0.971 | 0.912 | 0.838 | 0.873 | 0.913 | 18 | 9 |
| Decision Tree (RandomSearch) | 0.951 | 0.977 | 0.875 | 0.946 | 0.909 | 0.983 | 6 | 15 |
| Random Forest (default) | 1.000 | 0.997 | 1.000 | 0.973 | 0.986 | 1.000 | 3 | 0 |
| Random Forest (RandomSearch) | 0.998 | 0.999 | 0.991 | 1.000 | 0.996 | 1.000 | 0 | 1 |
| Random Forest (RandomSearch + GridSearch) | 0.998 | 0.999 | 0.991 | 1.000 | 0.996 | 1.000 | 0 | 1 |

## Threshold-based model selection (business goal: minimise FN)

| Model | Accuracy | Precision | Recall | F1 | F2 | AUC | FP | FN |
|---|---|---|---|---|---|---|---|---|
| Decision Tree (tuned) @thr=0.34 | 0.971 | 0.828 | 0.955 | 0.887 | 0.927 | 0.983 | 22 | 5 |
| Random Forest (tuned) @thr=0.38 | 0.992 | 0.941 | 1.000 | 0.969 | 0.988 | 1.000 | 7 | 0 |
| Random Forest (grid-tuned) @thr=0.43 | 0.996 | 0.965 | 1.000 | 0.982 | 0.993 | 1.000 | 4 | 0 |
| Random Forest (grid-tuned) @0.50 | 0.999 | 0.991 | 1.000 | 0.996 | 0.998 | 1.000 | 1 | 0 |
| LR + SMOTE @0.50 | 0.998 | 0.982 | 1.000 | 0.991 | 0.996 | 1.000 | 2 | 0 |


ROC curves — all models (test set)
True Positive Rate (TPR / Recall)
False Positive Rate (FPR)
0.0
0.2
0.4
0.6
0.8
1.0
LR baseline (AUC=1.000)
LR + Under-Sampling (AUC=1.000)
LR + Over-Sampling (AUC=1.000)
LR + SMOTE (AUC=1.000)
LR + ADASYN (AUC=1.000)
Decision Tree (default) (AUC=0.913)
Random Forest (default) (AUC=1.000)
Decision Tree (tuned) (AUC=0.983)
Random Forest (tuned) (AUC=1.000)
Random Forest (grid-tuned) (AUC=1.000)
Random (AUC=0.5)


Random Forest (grid-tuned)
default threshold 0.50
True label
Low Risk
High Risk
808
1
0
111
Low Risk
High Risk
Predicted label


Random Forest (grid-tuned)
business threshold 0.43
True label
Low Risk
High Risk
805
4
0
111
Low Risk
High Risk
Predicted label

## Conclusions & recommendations

*1. Best model & justification.* Random Forest (RandomizedSearch → GridSearch tuned, `class_weight='balanced'`, 500 trees, min_samples_leaf = 2) with a business threshold of 0.43. *On the held-out test set (920 customers, 111 high-risk) it catches all 111 risky customers (recall 1.00)*, with precision 0.965, F1 0.982, AUC-ROC 1.000. It beats the Decision Tree (recall 0.84–0.95) and matches Logistic Regression's recall while being more robust to non-linear patterns; the threshold is chosen from training-set out-of-fold predictions so the test set is untouched.

*2. Impact of imbalance handling. Majority-class prediction gives 87.9% accuracy but 0% recall, proving accuracy is misleading. For Logistic Regression the data is almost linearly separable, so Under-sampling, Over-sampling, SMOTE and ADASYN all kept recall at 1.00 and AUC at 1.00 but slightly lowered precision/F1 (1.00 → 0.98–0.99). For tree models, class weighting + tuning is what removed false negatives (Random Forest FN 3 → 0; Decision Tree FN 18 → 6). Take-away: resampling is a tool to* test*, not to apply blindly.

*3. Metric comparison.* See the summary table above: Logistic Regression variants (F1 0.98–1.00), default Decision Tree (F1 0.87, AUC 0.91), tuned Decision Tree (F1 0.91, AUC 0.98), tuned Random Forest (F1 0.996, AUC 1.00) and the final threshold-tuned Random Forest (F1 0.982, recall 1.00).

*4. Business interpretation.* False Negative (Type-II): *a risky customer is not flagged → future default or fraud loss, no early intervention. This is the expensive error.* False Positive (Type-I):* a safe customer is flagged → extra manual review and possible customer friction; a bounded, recoverable cost. Hence recall/F2 is prioritised and the threshold is lowered to 0.43; in our test set that costs 4 extra reviews out of 809 safe customers (0.5% FPR) in exchange for zero misses.

*5. Recommendations & caveats. (i) Deploy the Random Forest as a first-line alert, with the threshold reviewed periodically against actual review capacity. (ii) Monitor drift in late-payment days, missed payments and credit score — the top drivers. (iii) Caveat:* near-perfect scores suggest the dataset is synthetic and almost linearly separable; real portfolios are noisier, so expect lower recall/precision and re-validate on real data (time-based split) before production. (iv) Keep a human in the loop and check fairness across gender/region before using alerts to take adverse actions.